Ecology and Ecosystems

Energy flow, food webs, populations, and environmental balance

This material is designed as a concise student handout. It introduces core ideas, important vocabulary, and short review questions for classroom or self-study use.

Learning outcomes

- Explain the core concepts using accurate biology terminology.
- Connect biological structures to their functions.
- Apply the concepts to simple examples and review questions.
- Use the vocabulary section to strengthen scientific communication.

1. Ecosystem components

An ecosystem includes living organisms and the non-living environment they interact with. Living components include plants, animals, fungi, and microbes.

Non-living components include sunlight, water, air, soil, temperature, and minerals. Both parts are needed to understand how ecosystems function.

Key points

- Biotic factors are living parts of an ecosystem.
- Abiotic factors are non-living environmental parts.
- Organisms interact with each other and with their physical environment.

Mini activity

List three biotic and three abiotic factors in a school garden or local park.

2. Energy flow

Energy enters most ecosystems through producers such as plants and algae. Producers use photosynthesis to convert light energy into chemical energy.

Consumers obtain energy by eating producers or other consumers. Decomposers recycle nutrients by breaking down dead organisms and waste.

Key points

- Producers form the base of most food chains.
- Energy decreases at higher trophic levels.
- Decomposers recycle nutrients back into the ecosystem.

Mini activity

Create a food chain with at least four organisms and label producer, consumer, and decomposer.

3. Population interactions

Organisms interact through competition, predation, mutualism, parasitism, and commensalism. These relationships influence survival and population size.

A balanced ecosystem does not mean nothing changes. It means populations and resources fluctuate within limits that the ecosystem can support.

Key points

- Competition occurs when organisms need the same limited resource.
- Predation affects both predator and prey populations.
- Symbiosis describes close relationships between different species.

Mini activity

Describe one example of mutualism, such as bees pollinating flowering plants.

Important vocabulary

Term	Meaning
Ecosystem	Community of organisms plus their physical environment.
Producer	Organism that makes its own food, usually by photosynthesis.
Consumer	Organism that obtains energy by eating other organisms.
Decomposer	Organism that breaks down dead matter and waste.
Trophic level	Feeding position in a food chain or food web.

Quick review quiz

1. What is the difference between biotic and abiotic factors?
2. Why are producers important in ecosystems?
3. Why is energy transfer between trophic levels inefficient?
4. Give one example of competition in nature.
5. How do decomposers help maintain ecosystem health?

Teacher note: Students should answer the review questions before checking a textbook, notes, or class discussion.